Small Signal Stability

↓

Representation of the System

Any power system can be written as 'n' coupled first-order nonlinear ODEs.

state-space eqn $\dot{x} = f(x, u, t)$

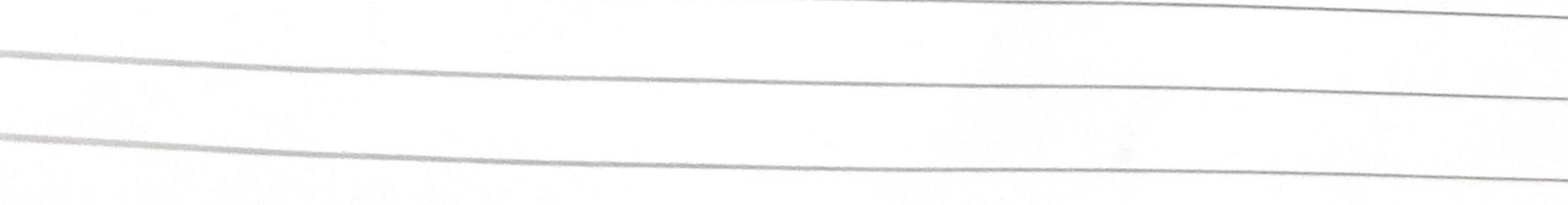
where x : is the state vector (angle, speed, flux, AVRs)

u : input vector (Tm, ref-Vol, ...)

f : is in general non-linear.

↓

the output eqn $y = g(x, u)$ maps states to measured quantities (terminal voltages, power flows).



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→ Equilibrium point

↓

An equilibrium is the operating point where nothing changes $\dot{x} = f(x_0, u_0) = 0$

↓

For a power System, this is a steady-state load flow Solution.

↓

X_0 = state Vector x at Equilibrium

↓

If 'f' is a Set of linear functions, the System is linear. linear systems have only one equilibrium state.

↓

If '-f' is a Set of non linear functions, the Systems is non-linear. It may have more than one equilibrium state.

→ Linearization:

↓

Since f' is non linear, we cannot directly apply eigenvalues analysis. So we perturb the system by a small amount from equilibrium.

$$x = x_0 + \Delta x,$$
$$u = u_0 + \Delta u$$

Expanding $f(x_0+\Delta x, u_0+\Delta u)$ using Taylor series, keeping only first order terms.

$$f_i(x_0, u_0) + \frac{\partial f_i}{\partial x_1}\Delta x_1 + \cdots + \frac{\partial f_i}{\partial x_n}\Delta x_n + \frac{\partial f_i}{\partial u_1}\Delta u_1 + \cdots + \frac{\partial f_i}{\partial u_r}\Delta u_r$$

Since $\dot{x}_{i0} = f_i(x_0, u_0)$

$$\Delta \dot{x}_1 = \frac{\partial f_i}{\partial x_1}\Delta x_1 + \cdots + \frac{\partial f_i}{\partial x_n}\Delta x_n + \frac{\partial f_i}{\partial u_1}\Delta u_1 + \cdots + \frac{\partial f_i}{\partial u_r}\Delta u_r$$

similarly: $\Delta y_j = \frac{\partial g_j}{\partial x_1}\Delta x_1 + \cdots + \frac{\partial g_j}{\partial x_n}\Delta x_n + \frac{\partial g_i}{\partial u_1}\Delta u_1 + \cdots + \frac{\partial g_i}{\partial u_r}\Delta u_r$

↓

The linearised form of above eqns will be

(state vector)

$$\Delta \dot{x} = A\Delta x + B\Delta u$$
$$\Delta \dot{y} = C\Delta x + D\Delta u$$

$$\Delta\dot{x} = A \cdot \Delta x + B \cdot \Delta u$$

↓

The 'A' matrix (state or plant matrix) is the Jacobian of 'f' with respect to x - all the partial derivatives $\frac{\partial f_i}{\partial x_j}$ evaluated at equilibrium.

↓

This is the most important matrix in small signal stability. For a Syn. machine it contains terms from the swing eq, day flux eqs, & governor - all the dynamics model

state vector (n) ↑ → i/p vector (r)

$$\Delta\dot{x} = A \cdot \Delta x + B \Delta u$$

$$\Delta y = C \Delta x + D \Delta u$$

o/p vector (m)

A → state/plant matrix of (n×n)
B → control/i/p matrix n×r
C → o/p matrix m×n
D → feed forward matrix (m×r)

The linearization is only valid for small perturbations

Taking Laplace transform of eq

$$s\Delta x(s) - \Delta x(0) = A\Delta x(s) + B\Delta u(s)$$

$$\Delta y(s) = C\Delta x(s) + D\Delta u(s)$$

Recurrence eqn

$$(SI-A)\,\Delta x(s) = \Delta x(0) + B\,\Delta u(s)$$

Hence $\Delta x(s) = (sI-A)^{-1}\,[\Delta x(0) + B\Delta u(s)]$

&

$$\Delta y(s) = C(sI-A)^{-1}\,[\Delta x(0) + B\Delta u(s)] + D\Delta u(s)$$

The poles of this transfer matrix - the values of s where $(sI-A)$ becomes singular are the eigenvalues of A, found from $\det(sI-A)=0$

↓

these eigenvalues $\lambda = \sigma \pm j\omega$. carry all the stability information. The real part σ gives the decay/growth rate; the imaginary part 'ω' gives oscillation frequency

$\dot{x} = \sin x$

$\Phi(IA-A)$

After Imeerismey my power System unto $\Delta x = A \Delta x$, we entract the physical meanmey from the matrix A

eigen Values
↓
eigen Vectors
↓
modal Iremsformaliom
↓
Jaerticepation factors

⇒ The eigenvalues of a matrix are given by the values of the scalar parameter λ for which there exist non-trivial solutions to follow -

$$A\phi = \lambda\phi \longrightarrow \text{Vector } (n \times 1)$$

To find eigenvalues

$$(A - \lambda I)\phi = 0$$

For non-trivial sol

$$|A - \lambda I| = 0$$

For any eigenvalue λ_i

$$A\phi_i = \lambda_i \phi_i$$

the eigen vector ϕ_i is the right eigenvector eq has the form

$$\phi_i = \begin{bmatrix} \phi_{1i} \\ \phi_{2i} \\ \vdots \\ \phi_{Ni} \end{bmatrix}$$

→ left vectors

Similarly, the n-row vector ψ_i which satisfies

$$\psi_i A = \lambda_i \psi_i, \quad \psi_i = [\psi_1 \ \psi_2 \ \dots \ \psi_N]$$

↓

For each eigenvalue λ_p, there are two eigen vectors, as they carry completely different info.

↓

the right eigenvectors ϕ_i $(n \times 1)$ satisfies $A\phi_i = \lambda_i \phi_i$. It describes the shape of mode i - which state variable move, & in what relative proportion & phase, when mode i is excited

The left eigenvector ψ_i $(n \times 1)$ satisfies $\psi_i A = \lambda_i \psi_i$. It describes the observability of mode i from each state - i.e how much each state variable contributes to exciting mode i from initial conditions

Linearised State Eq.

$$\det(A - \lambda I) = 0$$

Eigenvalues $\lambda_1, \lambda_2 \cdots \lambda_n$

$$\lambda = \sigma \pm j\omega$$

Right eigenvector ϕ_p

$$A\phi_p = \lambda_p \phi_p$$

Left eigenvector ψ_p

$$\psi_p \cdot A = \lambda_p \psi_p$$

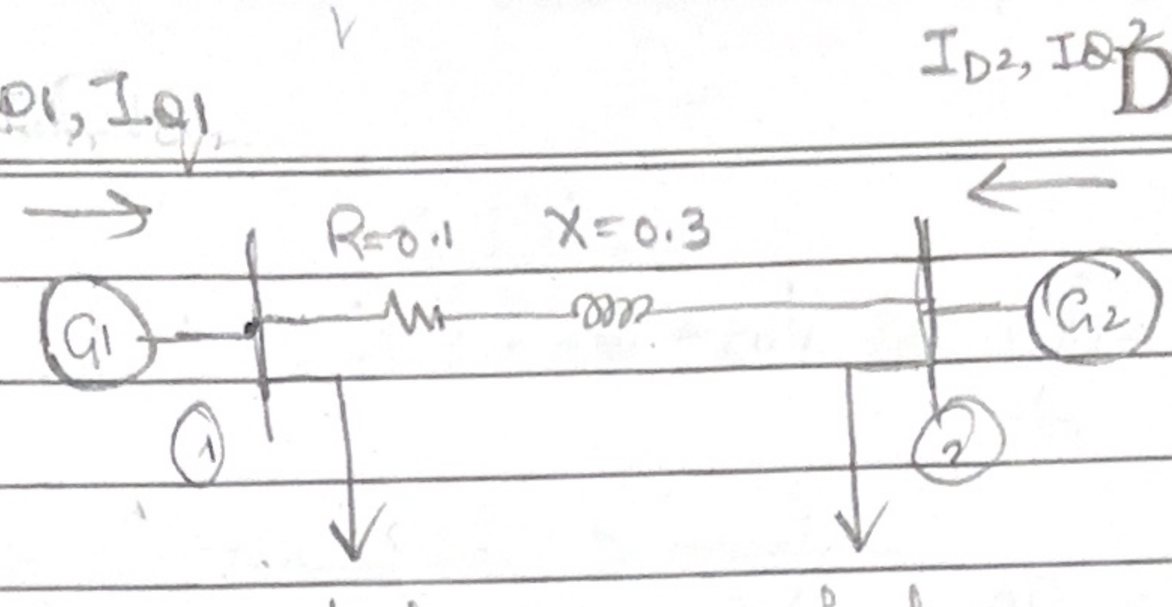
Modal decomposition $\underline{\Delta x = \phi z}$

$$\phi = [\phi_1, \phi_2, \cdots \phi_n], \phi^{-1} = \psi$$

↓

Time Response

$$\Delta x(t) = \sum C_p \phi_p e^{\lambda_p t}$$

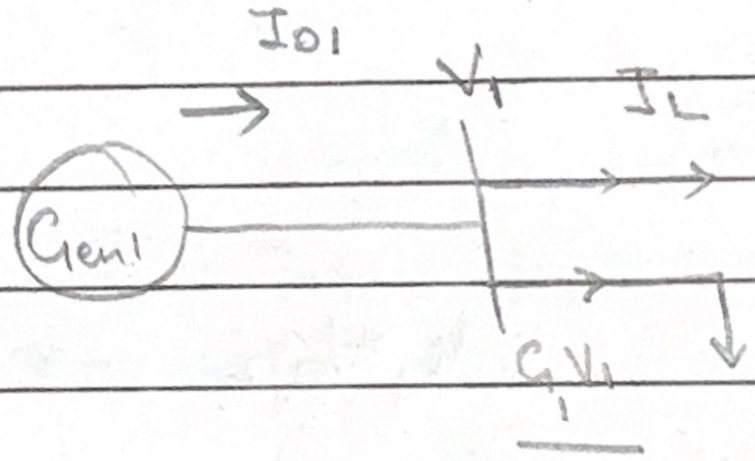


Network has V_{1D}, V_{1Q}, V_{2D}, V_{2Q}

& the line current I_{LD}, I_{LQ}

Machine char → $\delta_1, \omega_1, \psi_{d1}, \psi_{q1}, \psi_1 \ldots$

At Bus 1



KCL → $I_{G1} - G_1V_1 - I_L = 0$

$I_{D1} - G_1V_{1D} - I_{LD} = 0$

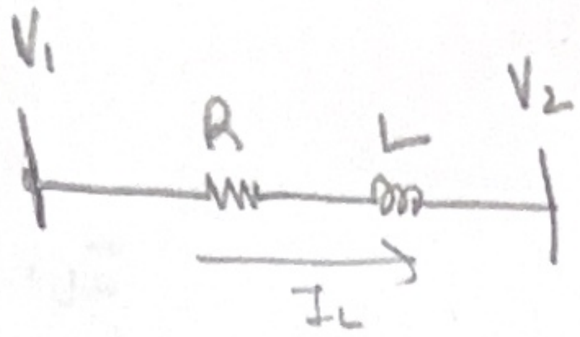
$I_{Q1} - G_1V_{1Q} - I_{1Q} = 0$

At Bus 2)

$I_{LD} - G_2V_{2D} - I_{D2} = 0$

$I_{LQ} - G_2V_{2Q} - I_{Q2} = 0$

Spiral



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KVL → $L\frac{d\dot{\gamma}_D}{dt} = V_{1D} - V_{2D} - R\dot{\gamma}_{LD} - X\dot{\gamma}_{LQ}$

$L\frac{d\dot{\gamma}_Q}{dt} = V_{1Q} - V_{2Q} - R\dot{\gamma}_{LQ} - X\dot{\gamma}_{LD}$

→ Generador 1)

↓

$\dot{\gamma}_{d1}, \dot{\gamma}_{q1}$

⇓ rotacion matem $\begin{bmatrix} I_{D1} \\ I_{Q1} \end{bmatrix} = C_p(\delta_1) \begin{bmatrix} I_{d1} \\ I_{q1} \end{bmatrix}$

I_{D1}, I_{Q2}

lly network Voltage

$V_{d1}, V_{q2} \rightarrow V_{D1}, V_{Q1}$

$M_x = A_x$

0 0 0 0 0 0 0 0 0 0 0 0	ΔV_{1D}	-G 0 0 0 -1 0 1 0 0 0		V_{1D}
0 0 0 0 0 G 0 0 0 0	ΔV_{1Q}	0 -G 0 0 0 G -1 0 1 0 0		V_{1Q}
0 0 0 0 0 0 0 0 0 0	ΔV_{2D} =	0 0 -G 0 1 0 0 0 -1 0		V_{2D}
0 0 0 0 0 0 0 0 0 0	ΔV_{2Q}	0 0 0 -G 0 1 0 0 0 -1		V_{2Q}
0 0 0 0 L 0 0 0 0 0	$\Delta \dot{\gamma}_{LD}$	-1 0 1 0 R X 0 0 0 0		$\dot{\gamma}_{LD}$
0 0 0 0 0 L 0 0 0 0	$\Delta \dot{\gamma}_{LQ}$	0 -1 0 1 -X R 0 0 0 0		$\dot{\gamma}_{LQ}$
	ΔI_{D1}			I_{D1}
	ΔI_{Q1}			I_{Q1}
	ΔI_{D2}			I_{D2}
	ΔI_{Q2}			I_{Q2}

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Stationary Ref to Rotating ref.

↓

Physical branches like. inductor, capacitor, or winding follow Kirchoff/Faraday's law in stat. frame

↓

eg $\frac{L\,dI_{abc}}{dt} = V_{abc} - R\,I_{abc}$

↓

In Rotating ref. frame, the network's frame (DQ) rotating at nominal freq. ω_b.

↓

To transform the stationary to rotating frame at angular frequency ω is done by multiplying a rotation matrix

↓

$$T(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$T(\theta)$ or

↑

$P(\theta), \theta = \int \omega \cdot dt$

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$$\vec{r}_{stat} = T(\theta)\,\vec{r}_{dqv}$$

$$\frac{d\vec{r}_{stat}}{dt} = \frac{d(T\vec{r}_i)}{dt}$$

uv-rule

$$= T\frac{d(\vec{r}_i)}{dt} + \dot{r}_i\frac{d(T)}{dt}$$

$$= T(\dot{r}_i) + \dot{T}(\vec{r}_i)$$

extra term comes when we convert/transform to rotating ref -

So what is this T or $\dot{T}$?

$$T = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$\rightarrow \quad \theta = \omega t$

$$\frac{d\theta}{dt} = \omega$$

$$\dot{T} = \omega\begin{bmatrix} -\sin\theta & \cos\theta \\ -\cos\theta & -\sin\theta \end{bmatrix}$$

$$\because \frac{d\cos\theta}{dt} = -\sin\theta\frac{d\theta}{dt}$$

$$= -\sin\theta\,\omega$$

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We know a special term ← Vector

$$J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

↓

multiply T by J, we get

$$\dot{T} = \omega T J$$

→ means $\dot{T}$ is also equal to $\omega T J$

↓

$$\dot{T} = \omega \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} T$$

→ -ve, depends on sign convention

eq $\left(\frac{d\lambda}{dt}\right)_{stat} = \left(\frac{d(i)}{dt}\right)_{rot} + \omega \underbrace{\begin{bmatrix} 0 & -i \\ \lambda & 0 \end{bmatrix}} \lambda$

↓

$$\underbrace{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} I_D \\ I_Q \end{bmatrix}} = \begin{bmatrix} -I_Q \\ I_D \end{bmatrix}$$

↓

Coupling terms ← means d-axis depend on q-axis current

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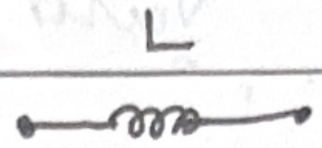
For Inductor



$$V = L\frac{di}{dt}$$

in-dq frame

$$V_D = L\dot{i}_D - L\omega I_q + R i_D$$
$$V_Q = L\dot{i}_Q + L\omega I_d + R i_q$$

$$X = \omega_0 L$$

$$\therefore V_D = L\dot{i}_D + R i_D - X I_q$$
$$V_Q = L\dot{i}_Q + X I_d + R I_q$$

For Capacitor

ully;

$$i = C\frac{dv}{dt}$$

dq

$$i_D = C\dot{V}_D - B V_Q$$
$$i_Q = C\dot{V}_Q + B V_D$$

$$B = \omega_b C$$

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↳ forming the Network Equations

↓

Inductor

$$\begin{bmatrix} V_D \\ V_Q \end{bmatrix} = L \begin{bmatrix} \dot{I}_D \\ \dot{I}_Q \end{bmatrix} + \begin{bmatrix} R & -X \\ X & R \end{bmatrix} \begin{bmatrix} I_D \\ I_Q \end{bmatrix}$$

Capacitor

$$i = C\frac{dv}{dt}$$

$\downarrow dv$

$$\frac{dv}{dt} = \dot{v} + \omega_b \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} v$$

Then,

$$\dot{i} = C\left[\dot{v} + \omega_b \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} v\right]$$

$$i = C\dot{v} + \omega_b C \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} v$$

$$\omega_b C = B$$

In dq- frame

↳

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = C \begin{bmatrix} \dot{v}_d \\ \dot{v}_q \end{bmatrix} + \begin{bmatrix} 0 & -B \\ B & 0 \end{bmatrix} \begin{bmatrix} V_D \\ V_Q \end{bmatrix}$$

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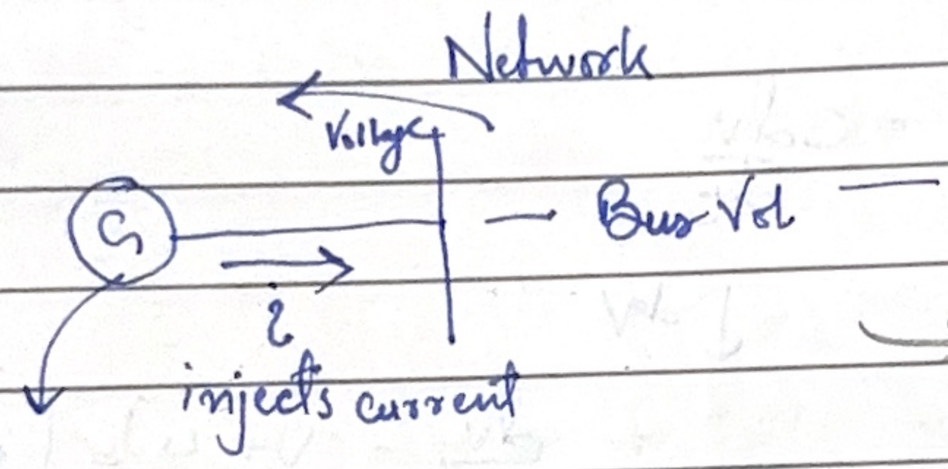
Combining Network & Machines Into one System

↓ 1) Network → $M_{net}\ddot{X}_{net} = \Delta_{net} X_{net}$

2) Generators → $M_{gen}\ddot{X}_{gen} = \Delta_{gen} X_{gen}$



$\psi_d, \psi_q, \psi, \omega, \delta$ ---

→ m/c inject Current

↓

Network Solves KCL

& determines terminal Vol

↓

that vol then affects m/c

↓ First Coupling

↳ Generator Current → Network

↓

m/c → $\Delta\psi_d$, $\Delta\psi_q$

dq- are m/c frame

↙ network doesn't understand

Bring to Common System

↓

DQ frame

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$$C_p(\delta_0) = \begin{bmatrix} \cos\delta_0 & -\sin\delta_0 \\ \sin\delta_0 & \cos\delta_0 \end{bmatrix}$$

Converts $\begin{bmatrix} I_d \\ I_q \end{bmatrix} \Rightarrow \begin{bmatrix} I_D \\ I_Q \end{bmatrix}$

$$\begin{bmatrix} I_D \\ I_Q \end{bmatrix} = \underbrace{C_p(\delta_0)} \begin{bmatrix} I_d \\ I_q \end{bmatrix}$$

↓ Rotation matrix

eg $\begin{bmatrix} I_d \\ I_q \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ all current due → along d-axis

↓ Rotate

$$\begin{bmatrix} I_D \\ I_Q \end{bmatrix} = C_p(\delta_0) \begin{bmatrix} I_d \\ I_q \end{bmatrix}, \quad \delta_1 = 30^\circ$$

$$C_p(\delta_1) = \begin{bmatrix} \cos\delta_0 & -\sin\delta_0 \\ \sin\delta_0 & \cos\delta_0 \end{bmatrix}$$

$$C_p(30^\circ) = \begin{bmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{bmatrix}$$

$$\begin{bmatrix} I_p \\ I_Q \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.866 \\ 0.5 \end{bmatrix}$$

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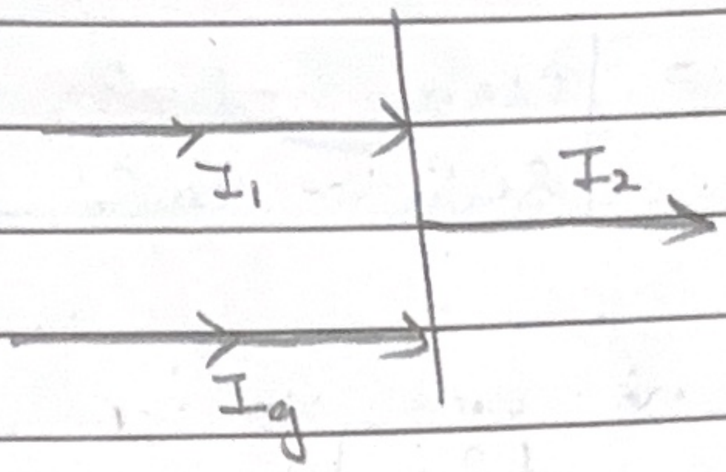
eg



Using KCL

$$I_1 + I_g - I_2 = 0$$

$$I_1 + I_D - I_2 = 0$$

ully for q-axis

Network Solver eq calculates

V_D, V_Q

$V_D = 1.02$

$V_Q = 0.12$

Now we rotate back ∵ gen = n don't use V_D, V_Q, they use V_d, V_q.

$$\begin{bmatrix} V_d \\ V_q \end{bmatrix} = C_p(\delta_1)^{-1} \begin{bmatrix} V_D \\ V_Q \end{bmatrix}$$

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∵ Rotation matrix is orthogonal

$$C_p^{-1} = C_p^T$$

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$$\Delta V_t = \frac{V_{sd}}{V_{to}} \Delta V_{gd} + \frac{V_{gq0}}{dt} \Delta V_{gq}$$

& AVR

↓ Rotating back to own dq-frame, in m/c. stator equations can compute new currents

stator

AVR

$$V_d = -R a_{2d} - \omega \psi_q + \frac{1}{\omega} \dot{\psi}_d$$

$$V_q = -R a_{2q} - \omega \psi_d + \frac{1}{\omega} \dot{\psi}_q$$

⇒ the generator & network continuously exchange infor

↳ Generator → gives Current

summary

↓

I_d, I_q

& network sees current & gives terminal Voltages

↳ V_d, V_q

But generator uses dq-axis
network uses DQ-axis

$$\begin{bmatrix} I_D \\ I_Q \end{bmatrix} = C_p(\delta_0) \begin{bmatrix} I_d \\ I_q \end{bmatrix}$$

① ②

↓ Because of this related Current depend on Current & rotor angle.

↓

meaning there are two independent reasons why network Current changes

1) Current changes
2) Rotor angle changes

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Suppose $y = Ax$

here $y = C\rho(\delta)x$

↓

the matrix itself depend on δ_0.

And $\delta \rightarrow$ itself or states

↙

∴ Both the vector & matrix change.

suppose $y = C(\delta)x$

Expand around equilibrium

$\delta = \delta_0 + \Delta\delta$

& $x = x_0 + \Delta x$

Now $C(\delta) = C(\delta_0 + \Delta\delta)$

$$C(\delta) \approx C(\delta_0) + \frac{dC}{d\delta}\Delta\delta$$

Increasing

$$\left(C_0 + \frac{dC}{d\delta}\Delta\delta\right)(x_0 + \Delta x)$$

$$= C_0x_0 + C_0\Delta x + \frac{dC}{d\delta}x_0\Delta\delta + \underbrace{\frac{dC}{d\delta}\Delta_y\Delta\delta}_{\text{zero}}$$

the perturbation becomes

$$\Delta y = C_0\Delta x + \frac{dC}{d\delta}x_0\Delta\delta$$

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Means

$$\Delta(\text{rotated quantity}) = C_p(\delta_0)\,\Delta(\text{quantity}) + \frac{dC_p}{d\delta}(\text{eq. qu.})\,\Delta\delta$$

current change resp. to due to rotation

the second term signifies

eg if current never changes $\Delta i = 0$
only rotor swings $\Delta\delta$

Captures . projection changes onto the network DQ-axis because m/c has rotated

$$\left(\frac{dC_p}{d\delta}\,i_0\,\Delta\delta\right)$$

independent

⇒ So there are two ways to inject current into the network.

1) Machine current itself changes because the machine dynamics changes → by $C_p(\delta_0)\Delta i$
Δi_d, Δi_q

2) Rotor angle changes, so same current vector from different orientation → $\Delta\delta$;
→ by $\frac{dC_p}{d\delta}\,i_0\,\Delta\delta$

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Initially

$$A_{sys} = \begin{bmatrix} A_{net} & 0 \\ 0 & A_{gen} \end{bmatrix}$$

the coupling terms are then inserted into off-diagonals

$$A_{sys} = \begin{bmatrix} A_{net} & \text{Net} \leftarrow \text{Mac} \\ \text{Mac} \leftarrow \text{Net} & A_{Gen} \end{bmatrix}$$

I $dq \rightarrow DQ$ I

Converts machine dq currents into network DQ current injections

Converts network DQ voltages into machine dq voltages & feeds them to stator flux eq

AVR

Date .....

To Connect machines with the network

↓

Ist Connection => (Current Connectivity)

↓

the m/c Computer currents

$$\begin{bmatrix} I_d \\ I_q \end{bmatrix}$$

But the network uses

$$\begin{bmatrix} I_D \\ I_Q \end{bmatrix}$$

$$\therefore \quad \begin{bmatrix} I_D \\ I_Q \end{bmatrix} = C_p(\delta) \begin{bmatrix} I_d \\ I_q \end{bmatrix}$$

Descripter => are written as

$$\begin{bmatrix} I_D \\ I_Q \end{bmatrix} - C_p(\delta) \begin{bmatrix} I_d \\ I_Q \end{bmatrix} = 0$$

This will be inserted in global matrix

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→ Linearising it

Network Current

$$\begin{bmatrix} I_D \\ I_Q \end{bmatrix} = \begin{bmatrix} I_{D0} \\ I_{Q0} \end{bmatrix} + \begin{bmatrix} \Delta I_D \\ \Delta I_Q \end{bmatrix}$$

Machine Current

$$\begin{bmatrix} I_d \\ I_q \end{bmatrix} = \begin{bmatrix} I_{d0} \\ I_{q0} \end{bmatrix} + \begin{bmatrix} \Delta I_d \\ \Delta I_q \end{bmatrix}$$

Set $\begin{bmatrix} I_d \\ I_q \end{bmatrix} = I_m$ $\qquad \begin{bmatrix} I_D \\ I_Q \end{bmatrix} = I_N$

So

$$I_m = I_0 + \Delta I$$

$$I_N = I_{N0} + \Delta I_N$$

Rotor angle $\longrightarrow$ $\delta = \delta_0 + \Delta\delta$

$$\therefore \quad (I_{N0} + \Delta I_N) = C_p(\delta_0 + \Delta\delta)\,(V_0 + \Delta V)$$

using Taylor Series

$$C_p(\delta_0 + \Delta\delta) \approx C_0 + C_\delta \Delta\delta$$

$$C_\delta = \left.\frac{dC_p}{d\delta}\right|_{\delta_0}$$

$$C_p(\delta_0) = C_0$$

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$$I_0 + \Delta I = (C_0 + C_\delta \Delta\delta)(i + \Delta\dot{\delta})$$

$$I_0 + \Delta I = C_0 i_0 + C_0 \Delta\dot{i} + C_\delta i_0 \Delta\delta + C_\delta \Delta i \Delta\delta$$

$\Rightarrow$ Ignoring second order terms i.e $\Delta i \Delta\delta$

Here ; $I_0 = C_0 i_0$

$$\therefore \quad \Delta I = C_0 \Delta\dot{i} + C_\delta i_0 \Delta\delta$$

$$\boxed{\Delta I - C_0 \Delta\dot{i} - C_\delta i_0 \Delta\delta = 0}$$

this is generator almost entire upper right off diagonal block.

→ Identifying every coefficient

$$\Delta I - C_0 \Delta\dot{i} - C_\delta i_0 \Delta\delta = 0$$

eg $x = \begin{bmatrix} \Delta I \\ \Delta\dot{i} \\ \Delta\delta \end{bmatrix}$ → Coefficient on

$\Delta I \longrightarrow +I$

$\Delta\dot{i} \longrightarrow -C_0$

$\Delta\delta \longrightarrow -C_\delta i_0$

Coeff. of $\Delta\dot{i}$ depends on rotation matrix

Coeff. of $\Delta\delta$ depend on derivative of rotation matrix

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→ To derive Voltage Coupling

↓

Connecting network to machines

Network → $V_n = \begin{bmatrix} V_D \\ V_Q \end{bmatrix}$, machine → $V_m = \begin{bmatrix} V_d \\ V_q \end{bmatrix}$

non-linear → $V_m = C_p^T(\delta) V_n$

or

$V_m = C_p^{-1}(\delta) V_n$

$$\boxed{V_m - C_p^T(\delta) V_n = 0}$$

Linearize:

$$V_n = V_0 + \Delta V$$

$$V_m = V_{m0} + \Delta V_m$$

$$\delta = \delta_0 + \Delta\delta$$

$$V_{m0} + \Delta V_m = C_p^T(\delta_0 + \Delta\delta)(V_0 + \Delta V)$$

$$V_{m_0} + \Delta V_m = C_0^T V_0 + C_0^T \Delta V + C_\delta^T V_0 \Delta\delta$$

$$\Delta V_m = C_0^T \Delta V + C_\delta^T V_0 \Delta\delta$$

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$$\Delta V_m - C_0^T \Delta V - C_\delta^T V_0 \Delta\delta = 0$$

$\Delta V_m \longrightarrow +I$

$\Delta V \longrightarrow -C_0^T$

$\Delta\delta \longrightarrow -C_\delta^T V_0$

These become lower-left off-diagonal block

the network can depend upon m/c current & rotor angle perturbation

$$A_{sys} = \begin{bmatrix} A_{net} & C_0 - C_\delta \dot{\delta}_0 \\ -C_0^T \quad -C_\delta^T V_0 & A_{gen} \end{bmatrix}$$

The m/c equations depend on network Voltage & rotor angle perturbation

$-C_p(\delta_0)$ from $I_n = C_p I_m$ → Converts dq → DQ

$-\frac{\partial C_p}{\partial \delta} i_0$ from diff. rotation matrix → Δ'δ' changes the direction of injected current even if i_d & i_q are constant

$-C_p^T(\delta_0)$ from $V_m = C_p^T V_n$ → DQ → dq

$-\frac{\partial C_p^T}{\partial \delta} V_0$ from diff Voltage rotation → Δ δ- changes how the m/c sees the same network

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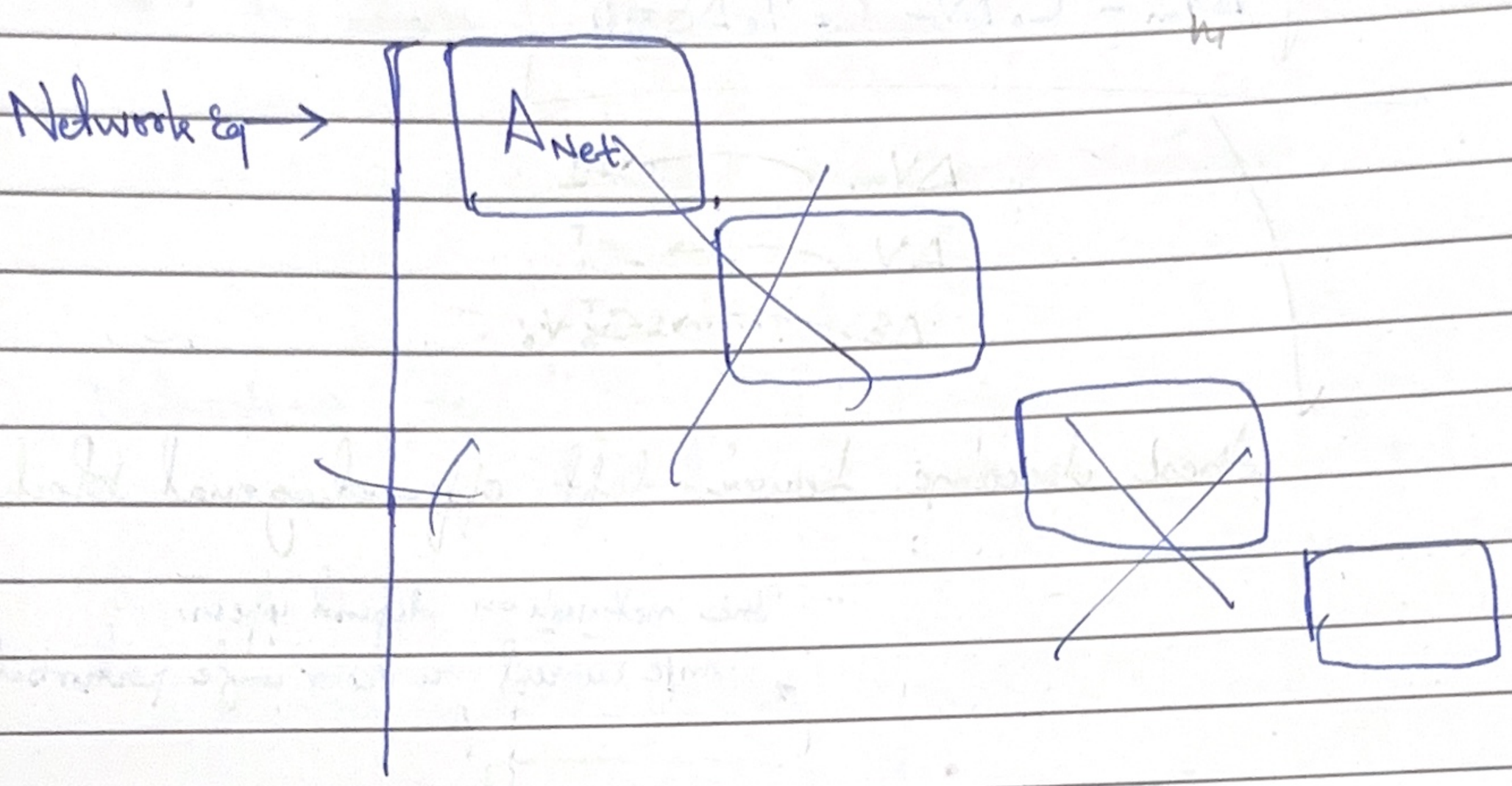

$$x = \begin{bmatrix} x_{net} \\ x_{g_1} \\ x_{g_2} \\ \Delta I_{DQ} \\ \Delta V_{dq} \end{bmatrix}$$

$$x_{net} = \begin{bmatrix} V_D, V_Q, I_D, I_Q, \cdots \end{bmatrix}^T$$

$$x_{g_1} = \begin{bmatrix} \delta_1, \omega_1, \psi_{fd_1}, \psi_{kd_1}, \psi_{kq_1}, \lambda_{d_1}, \lambda_{q_1}, E_{fd_1} \cdots \end{bmatrix}^T$$

↳ Interface Variable

$$\Delta I_{DQ} = \begin{bmatrix} \Delta I_D \\ \Delta I_Q \end{bmatrix}, \quad \Delta V_{dq} = \begin{bmatrix} \Delta V_d \\ \Delta V_q \end{bmatrix}$$

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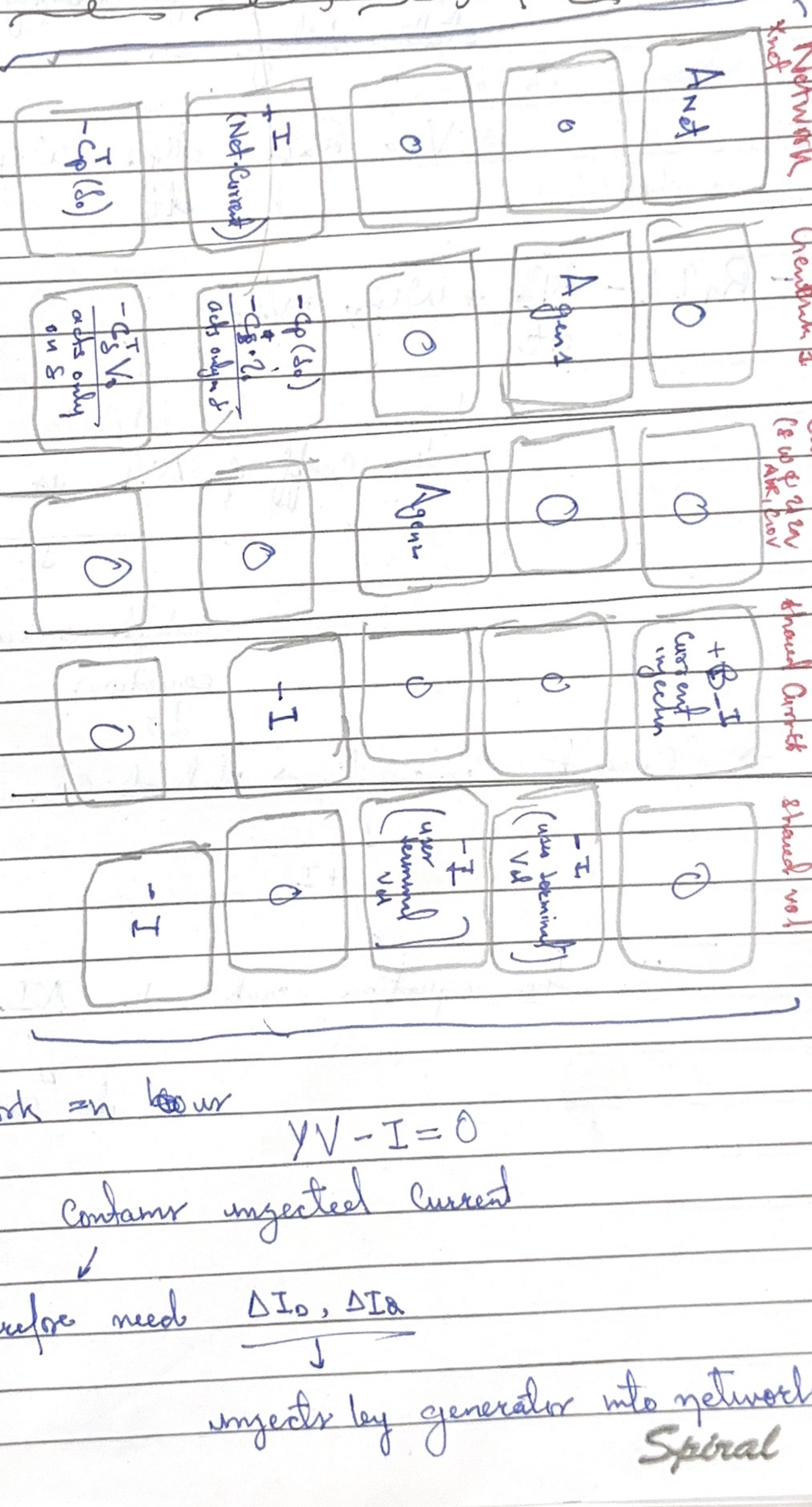
Network Eq

Generator 1 eq-

Generator 2 - eq

Current Continuity

Voltage Continuity



-Ks ΔVo

The network = n bus

$$YV - I = 0$$

Contains injected current

↓

Therefore need ΔI_D, ΔI_Q

↓

injects by generator into network

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→ Generator Eq '-I'

↓

Thus comes directly from stator voltage eq

$$V_d = R_a I_d + \frac{d\psi_d}{dt} - \omega \psi_q$$

$$-R_a I_d - \frac{d\psi_d}{dt} + \omega \psi_q + V_d = 0$$

↓

the coeff of ΔV_d is $-I$

↓

which is not a coupling equation

→ Current Continuity → Network Column

↓

+I

↓

∴ The equation starts with ΔI_{nd} - ... =0

↓

its coeff is +I

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→ Current Continuity row → Generator down

↓

Two diff matrices

$-C_p(\delta_0)$ &

↓ acts on only

$\Delta\delta_d$ $\Delta\delta_q$

↓

these come from $C_p(\delta_0)\Delta i$

$-C_{p\delta_0}$ or $-\frac{dC_{p\delta_0}}{d\delta}$

↓ acts only on

$\Delta\delta$

↓

it comes from $\frac{dC_p}{d\delta}\delta_0 \Delta\delta$

→ Voltage row → Network Column of

$-C_p^T$

↳ $\Delta V_0, \Delta V_0$

& in Generator Column

$-C_s^T V_0$

↳ $\Delta\delta$

$YV - I = 0$

$A_{net} x_{nd} + B_1 \Delta I = 0$

↳ $\Delta I_D \Delta I_Q$

δv

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	X_{net}	X_{gen}	$\Delta I_D \Delta I_Q$	$\Delta V_d \Delta V_q$
Network eq →	A_{net}	0	BI	0
Gen eq →	0	A_{gen}		$-I$
Current Continuity	I	$-C_P \cdot C_S \dot{z}_i$	$-I$	0
Vol continuity	$-C_P^T$	$-C_P^T V_0$	0	$-I$

Gen counter current

Network counter Vol

Current Path

z_d, z_q

↓

$-C_P, -C_S \dot{i}_0$

↓

$\Delta I_D, \Delta I_Q$

↓

BI

↓

Network eq

Voltage path

V_D, V_Q

↓

$-C_P^T, -C_S^T \cdot V_0$

↓

$\Delta V_d, \Delta V_q$

↓

$-I$

↓

Generator equation

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↳ Stator Transients

↓

Sync. machine stator winding is electrically a resistor in series with time varying flux-linked coil.

↓

By Faraday's law & Kirchoff's Voltage Law

↓ For phase a. $V_a = -R_a i_a - \frac{d\psi_a}{dt}$

↳ same for phase b & c

$\therefore V_{abc} = -R_a i_{abc} - \frac{d\psi_{abc}}{dt}$

$\frac{d}{dt}$

↳ transformer Voltage

↓

↳ stator transient term

↓

the emf induced by rate of change of flux linkage.

→ From stationary to rotating - Clarke's

↓ In stat. frame → $\vec{\psi}_s(t) = \psi_\alpha(t) + j\psi_\beta(t)$

↓ the dq-rotor frame → $\vec{\psi}_{dq}(t) = \vec{\psi}_s(t) e^{-j\theta(t)}$

$\vec{\psi}_s(t) = \vec{\psi}_{dq}(t) e^{j\theta(t)}$

$\vec{\psi}_{dq}(t) = \psi_d + j\psi_q$

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$$\vec{\psi}_s(t) = \psi_{dq} e^{j\theta(t)}$$

↓

Differentiating — (ov-Euler)

$$\frac{d\vec{\psi}_s}{dt} = \frac{d\psi_{dq}}{dt} e^{j\theta} + \vec{\psi}_{dq} \frac{d}{dt}(e^{j\theta})$$

$$\text{Since} = \frac{d}{dt} e^{j\theta} = j\theta e^{j\theta} = j\omega e^{j\theta}$$

→ Transport Theorem

$$\therefore \frac{d\vec{\psi}_s}{dt} = \left(\frac{d\vec{\psi}_{dq}}{dt} + j\omega \vec{\psi}_{dq} \right) e^{j\theta}$$

↓

the rate of change in the stationary frame equals the rate of change as measured in rotating frame. plus term → $j\omega\vec{\psi}_{dq}$

↓

∵ because frame rotates itself

↓

Voltage & Current transforms identically to flux

$$\vec{v}_s = \vec{v}_{dq} e^{j\theta}, \quad \vec{i}_s = \vec{i}_{dq} e^{j\theta}$$

As $$v_s = -R_a \vec{i}_s - \frac{d\vec{\psi}_s}{dt}$$

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$$\therefore \vec{V}_{dqv} e^{j\theta} = -R_a \vec{i}_{dqv} e^{j\theta} = \left(\frac{d\psi_{dqv}}{dt} + j\omega \psi_{dqv} \right) e^{j\theta}$$

$$\vec{V}_{dqv} = -R_a i_{dqv} - \frac{d\psi_{dqv}}{dt} - j\omega \psi_{dqv}$$

$\vec{V}_{dqv} = V_d + jV_q$, $\vec{i}_{dqv} = i_d + j i_q$

$\vec{\psi}_{dqv} = \psi_d + j\psi_q$ $-j\omega(\psi_d + j\psi_q) = \omega\psi_q - j\psi_d$

The stator voltage equations in dq-

$$V_d = -R_a i_d - \dot{\psi}_d + \omega \psi_q$$

$$V_q = -R_a i_q - \dot{\psi}_q + \omega \psi_d$$

$\omega\psi_q$ → speed Vol due the frame rotation

$R_a i_q$ → resistive drop

$\dot{\psi}_q$ → transformer voltage or Stator transient

i_d, i_q → used in Current Continuity

$$\begin{bmatrix} I_D \\ I_Q \end{bmatrix} = C_p(\delta) \begin{bmatrix} I_d \\ I_q \end{bmatrix} \quad \text{or} \quad \Delta I - C_p \Delta i - \frac{\partial C_{pi}}{\partial \delta} \Delta\delta = 0$$

$$YV - I = 0 \rightarrow A_{net} K_{net} + B_I \Delta I = 0$$

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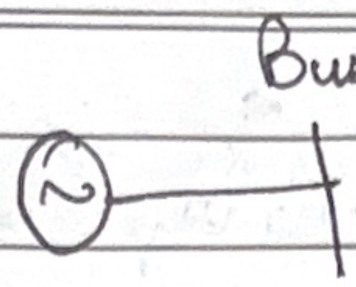
Suppose



$$I_K = \sum_j Y_{kj} V_j$$

if generator is connected

$$I_{gen} + \sum_j Y_{kj} V_j = 0$$

I_D, I_D

→ AVR

↳ AVR doesn't use V_d & V_q directly instead it uses the terminal voltage magnitudes

$$V_t = \sqrt{V_d^2 + V_q^2}$$

Compute error

$$e = V_{ref} - V_t$$

AVR dynamics solves

$$T_A \dot{E}_{fd} = K_A (V_{ref} - V_t) - E_{fd}.$$

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$\Delta\psi_F$, $\Delta\psi_G$ → transient flux delay

$\Delta\psi_H$, $\Delta\psi_K$ → Subtransient flux delay

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→ Network supplies

↓ V_D, V_Q

↓ Initial values/info available to machine

↓

$\delta, \omega, \psi_F, \psi_G, \psi_H, \psi_K, \psi_d, \psi_q, E_{fd}, X_g, X_e$

↓ At receiver V_D, V_Q

↓ transforms

V_d, V_q

↓

① Calculate stator current

$0 = C_1\psi_F + C_3\psi_4 - \psi_{dq} + X''_d I_{dq}$

② The stator equations

↓

$V_d = R_a I_d + \frac{1}{\omega_b}\dot{\psi}_d - \omega\psi_q$

↓

$\dot{\psi}_d = \omega_b\left(V_d - R_a I_d + \omega\psi_q\right)$

↓

m/c known $\dot{\psi}_{dq}$

③ Solver damper eq field eqn

i) $T'_d\dot{\psi}_F = -\psi_F + \psi_H + C_1 E_{fd}$ → field winding

ii) $T'_q\dot{\psi}_G = -\psi_G + \psi_K$ → G winding

iii) $T''_d\dot{\psi}_H = -\psi_H + \psi_d$ → H-winding

iv) $T''_q\dot{\psi}_K = -\psi_K + \psi_q$ → K-winding

M/C → known $\dot{\psi}_F, \dot{\psi}_G, \dot{\psi}_H, \dot{\psi}_K$

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(4) Calculate the electromagnetic torque

$$T_e = \psi_d i_q - \psi_q i_d$$

↓ machine knows T_e

(3) Solve AVR

↓

$$T_a \dot{E}_{fd} = -E_{fd} - K_a V_t$$

↓

knows now $\dot{E}_{fd}$

(2) Solver Turbine

$$\dot{X}_e = \omega_b X_{qv} - \frac{T_{mo}}{\omega_b} w$$

(1) Solver Swing eq

$$\frac{2H}{\omega_b} \dot{w} = T_m - T_e - Dw$$

(8) Solver rotor-angle

$$\dot{\delta} = w$$

(11) Integrating all States

$$x(t+\Delta t) = x(t) + \Delta \dot{x}$$

Spiral

For the Network

Date .....

→

↓

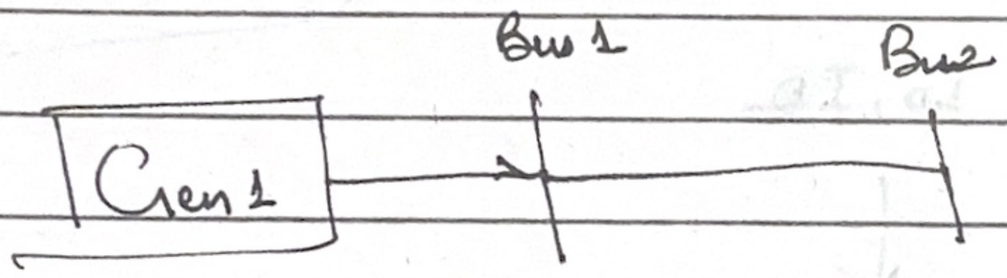
the machine solves the flux-current ⇒ ns

↳ i_d, i_q

$$i_d = \frac{\psi_d - C_1\psi_F - C_3\psi_4}{x''_d}$$

$$i_q = \frac{\psi_q - C_2\psi_G - C_4\psi_4}{x''_q}$$

$$\begin{bmatrix} i_D \\ i_Q \end{bmatrix} = C_p \begin{bmatrix} I_d \\ I_q \end{bmatrix} \qquad C_p = \begin{bmatrix} \cos\delta & -\sin\delta \\ \sin\delta & \cos\delta \end{bmatrix}$$



the Bus 1 eqn are

$$I_{D1} - G_1 V_{1D} - i_{LD} = 0$$

$$I_{Q1} - G_1 V_{1Q} - i_{1Q} = 0$$

↓ network already knows

G_1, i_{LD}, i_{LQ}

the machine supplies

I_{D1}, I_{Q1}

then network can solves for

V_{1D}, V_{1Q}

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C_1, C_2, C_3, C_4 → flux coefficient

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Forming the machine Equations

↓

m/c model → 2.2 → Subtransient model

Two rotor circuits per field (field F, damper H on d-axis; damper G & K on q-axis)

↓

Linearised state Vector per m/c → 13 states per m/c (with AVR & Governor)

Complete state Vector

$$\Delta X_{gen} = [\Delta\delta, \Delta\omega, \underbrace{\Delta\psi_F, \Delta\psi_G, \Delta\psi_H, \Delta\psi_K, \Delta\psi_d, \Delta\psi_q}_{\text{stator flux}}, \Delta I_d, \Delta I_q, \Delta E_{fd}, \underbrace{\Delta X_g, \Delta X_e}_{\text{governor coupled term}}]^T$$

d-axis, q-axis, AVR

$\Delta\dot{\delta} = \Delta\omega$ → rotor angle

$\Delta X_e = \frac{\omega_0}{\omega_0}\Delta X_g - \frac{T_{mo}}{\omega_0}\Delta\omega$ → $(\Delta X_e = \Delta T_m)$

→ $\frac{2H}{\omega_b}\Delta\dot{\omega} = -D\Delta\omega - \Delta T_e + \Delta T_m$ → ΔT_e from T_e → $(\psi_d I_q - \psi_q I_d)$

H → Inertia constant

→ Rotor flux → Phasor

→ $T'_d \Delta\dot{\psi}_F = -\Delta\psi_F + \Delta\psi_H + C_3 \Delta E_{fd}$ → Transient flux, d-axis

→ $T'_q \Delta\dot{\psi}_g = -\Delta\psi_g + \Delta\psi_K$ → Transient flux, q-axis damper

→ $T''_d \Delta\dot{\psi}_H = -\Delta\psi_H + \Delta\psi_d$

→ $T''_q \Delta\dot{\psi}_K = -\Delta\psi_K + \Delta\psi_q$ → Subtransient (d & q-axis)

→ stator flux

→ $\frac{1}{\omega_0}\Delta\dot{\psi}_d = -\omega_0\Delta\psi_q - R_a\omega_0\Delta I_d - \omega_0\Delta V_{gd}$

→ $0 = C_1\Delta\psi_F + C_2\Delta\psi_H - \Delta\psi_d + X''_d\Delta I_d$ → flux Current relation

→ $T_a\Delta\dot{E}_{fd} = -\Delta E_{fd} - K_a\Delta V_t$ → AVR, first order lag

→ $\left(\frac{T_{g1}}{R_{droop}}\right)\Delta\omega + \omega_0 T_{g2}\Delta X_g = \frac{-\Delta\omega}{R_{droop}} - \omega_0 X_g$ → governor droop